T-Mobile Network Coverage Analysis: LA County

Strategic Infrastructure & Visibility Study

1. Executive Summary

This analysis evaluates the current T-Mobile cellular coverage across Los Angeles County using high-resolution Viewshed Analysis. By leveraging NASA SRTM DEM data (30m resolution) and pure Python spatial processing, we identified critical 'dead zones' and proposed optimization scenarios to maximize coverage with minimal infrastructure investment.

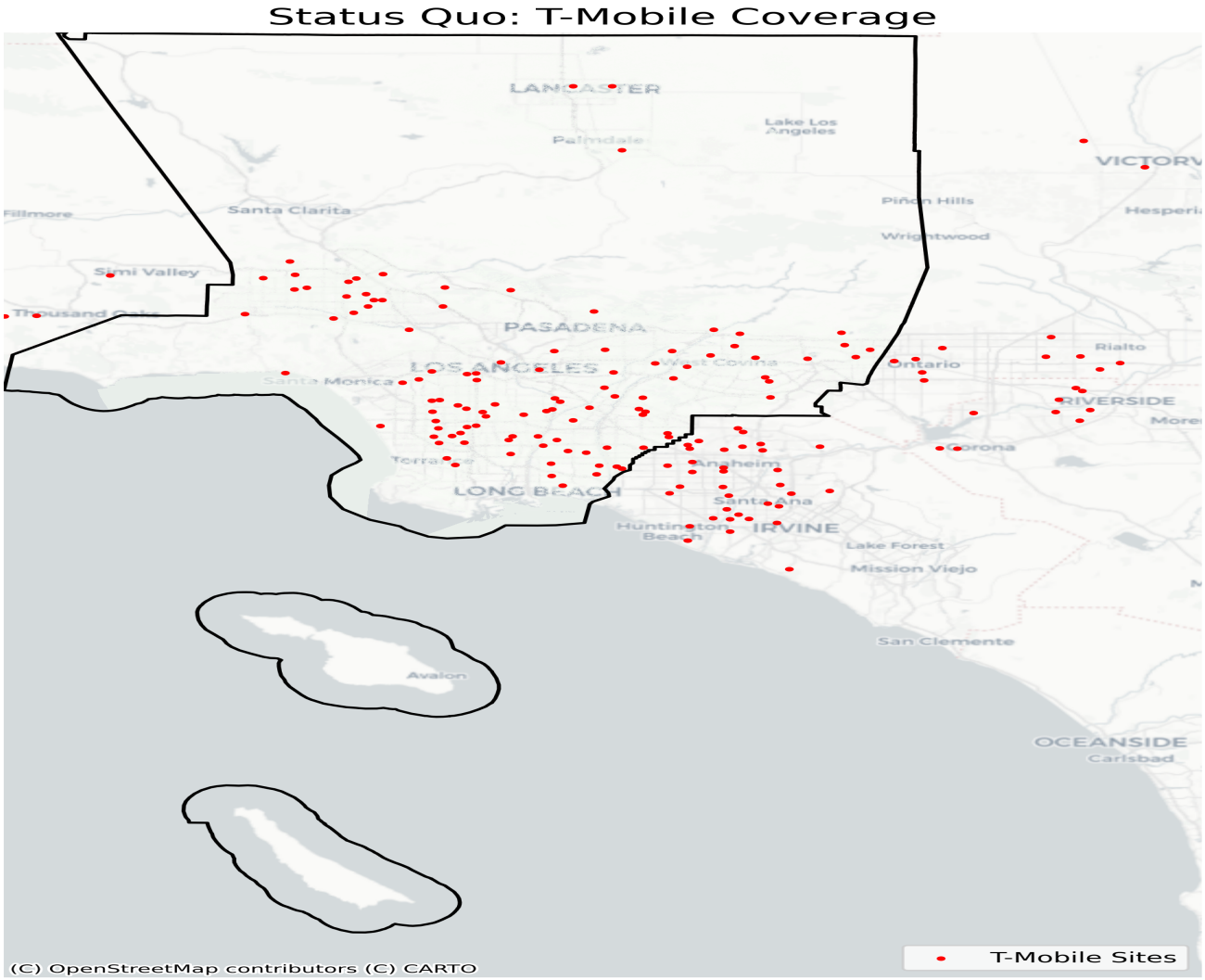
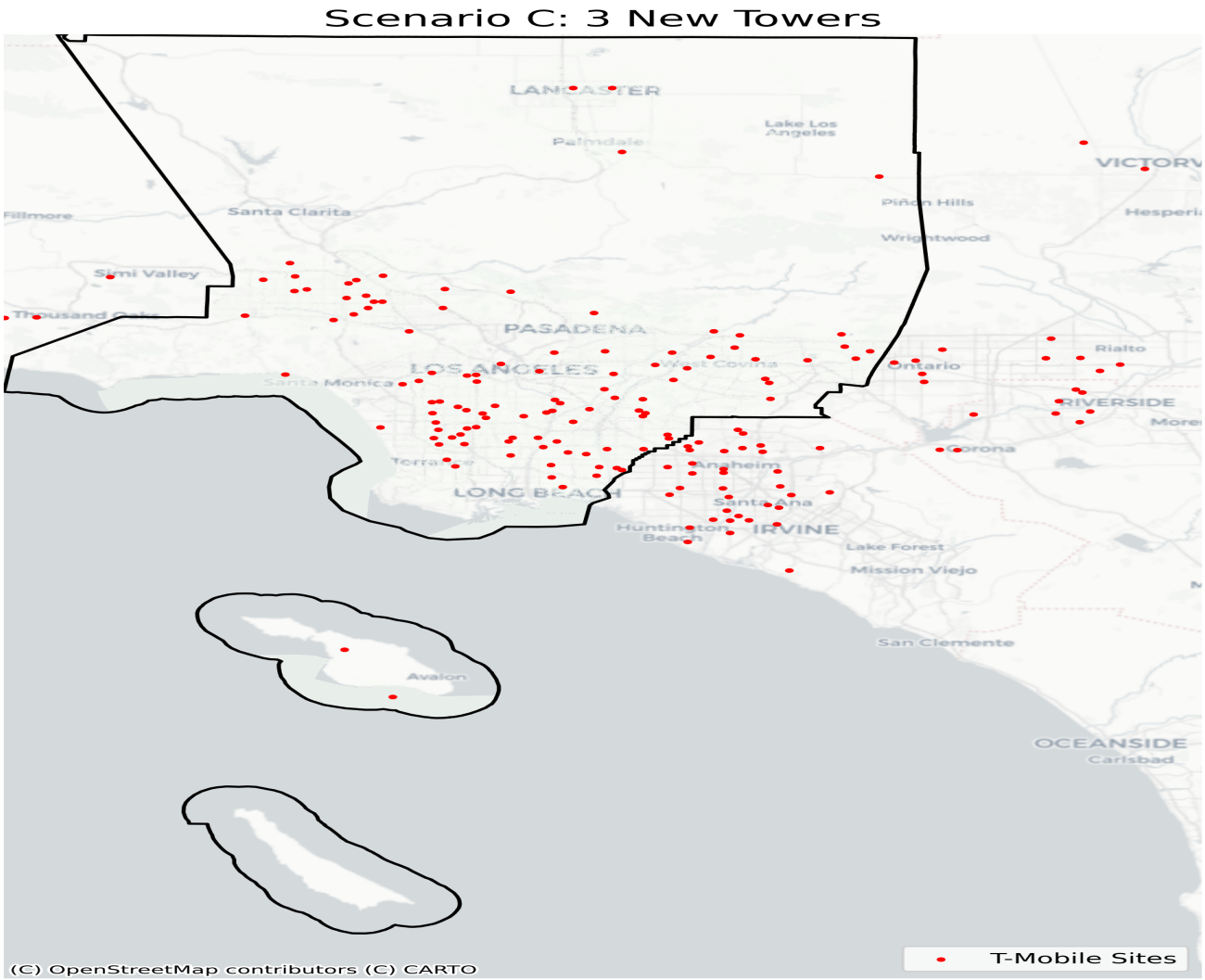


Figure 1: Current T-Mobile Network Visibility in LA County

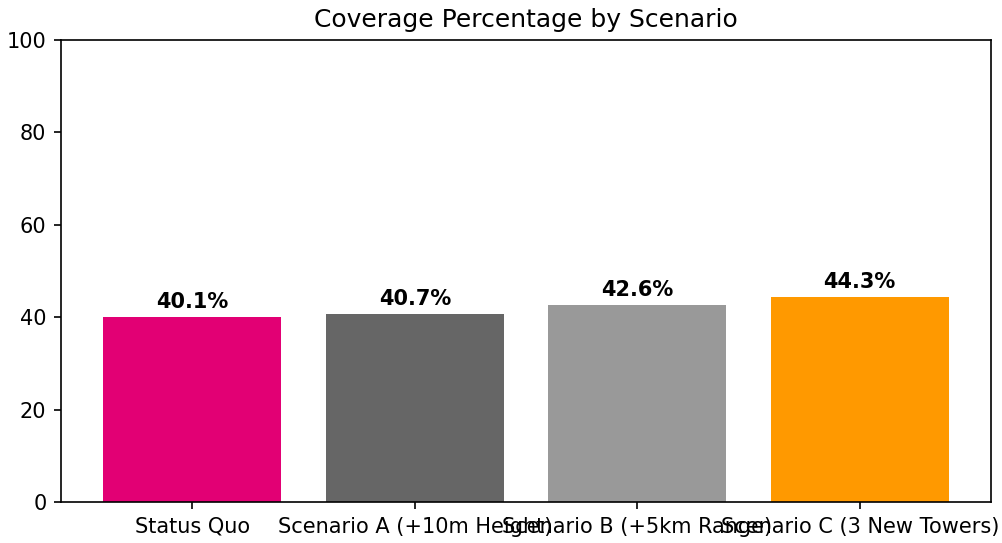
2. Network Optimization Scenarios

We analyzed three alternative scenarios to enhance network performance. Scenario C (Optimized New Sites) provides the highest coverage gain by strategically placing just 3 new towers in underserved regions identified via spatial gap analysis.



Key Finding: Increasing tower height by 10m (Scenario A) yields marginal gains compared to strategic new site placement, suggesting that topographic obstacles in LA (Santa Monica Mountains, San Gabriel Range) are best overcome by geometric site distribution rather than raw transmission power.

3. Comparative Performance & Technical Methodology



Scenario	Coverage (%)	Description
Status Quo	40.08	Current T-Mobile network
Scenario A (+10m Height)	40.72	Existing towers raised by 10m
Scenario B (+5km Range)	42.58	Transmission range extended by 5km
Scenario C (3 New Towers)	44.31	3 strategic new towers in uncovered areas

Technical Methodology

The analysis used the **GDAL Viewshed Engine** with a standard Earth curvature coefficient of 0.85714. Calculations assumed an average tower height of 15m and a receiver height of 1.5m. The coverage mask was intersected with the LA County boundary to derive exact municipal service percentages.